

As you know that electricity is one of the sources of energy. Electricity has become the basic need of today's life. Think about life without electricity?

In this Chapter you will learn about:

- Flow of current (direction).
- Types of electric circuits (parallel and series circuits).
- Energy transfer in an electric circuit.
- Effects of an electric current (heating, chemical, magnetic effects).
- Safety precautions: fuses and earth, the three pin plug, household circuits, KW hour and charges for electricity.
- Measuring current voltage and resistance (ammeter, voltmeter and resistors).
- Electricity at home.
- Electricity and safety (MCB, ELCB, earth wires)

All the students will be able to:

- ✓ Define current.
- ✓ Make parallel and series circuits.
- ✓ Investigate about types of circuits used for different purposes.
- ✓ Identify a disadvantage of a series circuit.
- ✓ Differentiate between current and energy.
- ✓ Explain the effects of electric current in daily use appliances.
- ✓ Describe voltage.
- ✓ Explain the resistance as an opposition to the flow of current.
- ✓ Describe the relationship between voltage and resistance.
- ✓ Measure current by using different devices.
- ✓ List the major uses of electricity in home.
- ✓ List Electrical hazards and precautionary measures to ensure the safe use of electricity at home.
- ✓ Describe why electricity is dangerous to humans.



Figure 11.1 Electrical Toaster



Figure 11.2 Electrical wires



Figure 11.3 LCD Display Monitor, TV

CURRENT

✓ Define current

The rate of flow of charge at certain point is called current.

Electrons of outermost shells of a conductor are capable of electricity flow. The electrons that have been knocked out of the outermost shell of an atom are known as free electrons. The movement of these free electrons constitutes an electric current which is the flow of electric charges. Electric current is measured in amperes (A). It is the amount of charge passing the given point in one second.

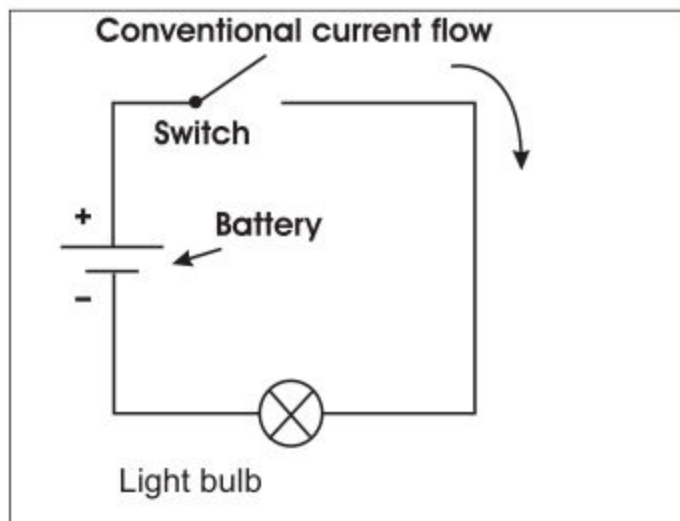


Figure 11.4 Electric current flows from one pole to the other of an electrical source

TYPES OF ELECTRIC CIRCUITS

✓ Make parallel and series circuits.

There are two main types of electric circuits, i.e. series circuits and parallel circuits.

1. Series Circuits:

If all the components are connected one after another in a single loop, then it is a series circuit. Current flows through a single path in a series circuit (figure 11.5). In this type of circuit, the amount of current which flows through each component (bulb) is the same.

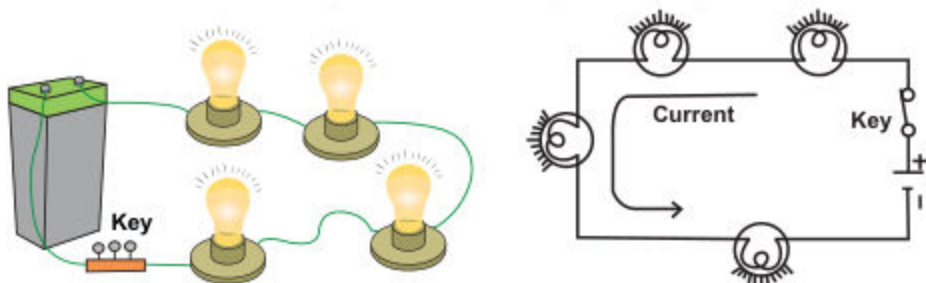


Figure 11.5 A series circuit provides only one path for the flow of current

Activity 11.1: Make a series circuit

What I need:

- Insulated wire (15 cm long 6 pieces).
- A 4.5 volt battery.
- 4.5 volt light bulbs in bulb holder (Four).
- A screw driver.
- A simple switch.

What to do:

- Connect the battery, switch and bulbs in a single loop with insulated wire as shown in the Fig 11.5.
- Add another light bulb and observe the brightness of bulbs.
- Replace one bulb with a burn out or fused bulb.

What I observed:

Activity Questions:

1. What happens to the brightness of the light bulbs when one bulb is added in the loop?
2. What happens when you replaced a light bulb with a fused bulb?

What I conclude:

2. Parallel Circuits:

If the components are connected in two or more loops, then it is a parallel circuit. Parallel circuits have more than one path for the current to flow (figure 11.6). The current flowing through different branches of a parallel circuit may be the same or different. But the current in each branch is less than the total current flowing out from the electrical source (battery).

As there is more than one path for the current to flow, a break in any branch of the circuit cuts the current-flow through that branch only.

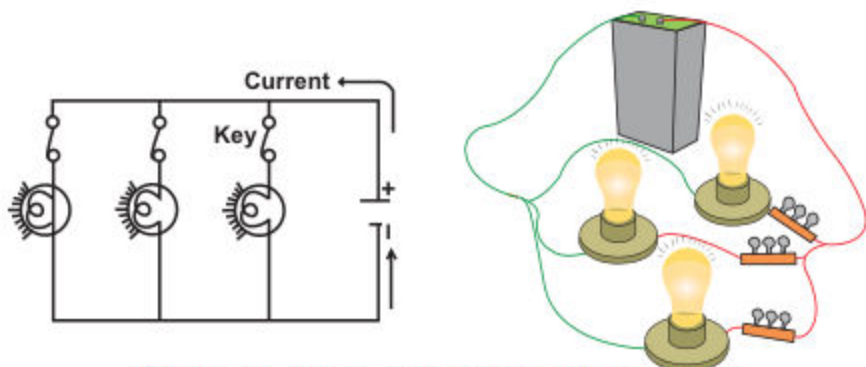


Figure 11.6 In a parallel circuit, there are more than single path for the current to flow.

Activity 11.2: Construct a parallel circuit.

What I need:

- Insulated wire (15 cm long 7 Pieces).
- A 4.5 volt battery.
- Three 4.5 volt light bulbs with bulb holder.
- A screw driver.
- A simple switch.

What to do:

1. Make a simple circuit with one bulb holder, the battery and the switch.
2. Connect a second bulb in parallel with the first bulb.
3. Add third bulb to the circuit parallel to the two bulbs.
4. Replace first bulb with a fused bulb and observe.
5. Replace the second bulb with a fused bulb and observe.

What I Observed:

Activity Questions:

1. What will happen if you unscrew one bulb from its holder?
2. Does the same thing happen if you replace one bulb with the fused bulb?

What I concluded:

USES OF CIRCUITS

- ✓ Investigate about types of circuits used for different purposes.

How many things around you right now use electric current? Current is used to transfer energy to many things. Each device that uses current is a part of at least one circuit, the circuit that supplies its voltage. Most electrical appliances have many circuits inside of them that are designed to carry out specific functions. Those circuits may be designed to light bulbs, move motor parts, or calculate. Each of those circuits may have thousands or even millions of parts.

DO YOU KNOW?

Thomas Edison invented the electric bulb in 1879. In 1880, he developed the first power plant to distribute the electricity in New York.

A series circuit uses a minimum amount of wire. However, a disadvantage of a series circuit is that all of the elements must be in working order for the circuit to function. The circuits in most business places and homes are connected in parallel. Look at the illustration of the kitchen and its wiring (figure 11.7). This is a parallel circuit, so even if one electrical device is switched off, the others can still be used. The circuits within many electrical devices are combinations of series



Figure 11.7 Parallel Circuit in Kitchen

circuits and parallel circuits. For example, a parallel circuit may have branches that contain several elements arranged in series.

DISADVANTAGE OF THE SERIES CIRCUIT

- ✓ Identify a disadvantage of a series circuit.

As there is only one path for the current to flow, a break at any part of the circuit stops the flow of current in the whole circuit; for example if one of the light bulbs burns out, the circuit will be broken and the other bulb will be dark, too. Series circuits have another disadvantage. Light bulbs and other resistors convert some energy into heat and light. The more light bulbs that are added to a series circuit, the less current there is available, and the dimmer all of the bulbs become.

DIFFERENCE BETWEEN CURRENT AND ENERGY

- ✓ Differentiate between current and energy.

Energy is ability to do work. It can neither be created nor be destroyed. It can be stored or transferred from one place to another or from object to object in different ways. Energy has different forms.

Electricity is one of the most useful and important forms of energy that can be transformed into many other forms of energy. Energy due to flow of electric charges within a circuit is called current electricity. We can say that current is a kind of energy.

EFFECTS OF ELECTRICAL CURRENT IN DAILY USE APPLIANCES

- ✓ Explain the effects of electric current in daily use appliances.
- ✓ List the major uses of electricity in homes.

Different devices are used for the conversion of electrical energy into other forms of energy, e.g. a light bulb transforms electrical energy to light energy. Electric bells and stereo players transform electrical energy to sound energy. A heater, hair dryer, electric iron and electrical toaster give us heat by using electrical energy while electric motors are used in a washing machine, vacuum cleaner,

electric drill and a fan which converts electrical energy into mechanical energy (figure 11.8 and 11.9).



Figure 11.8 Fan converts electrical energy into mechanical energy.

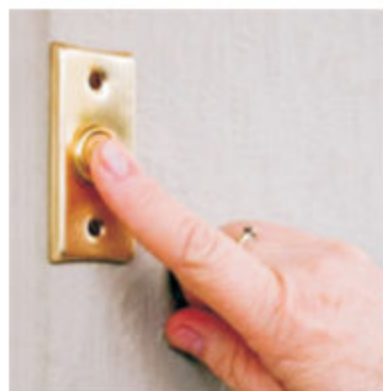


Figure 11.9 Door bell converts electrical energy into sound energy.

MEASURING CURRENT VOLTAGE AND RESISTANCE

✓Describe voltage.

Difference of potential between two points in a circuit or battery is called potential difference or voltage. Potential difference causes the charges to move through the conductor such as a wire. Charges will flow as long as there is a potential difference between the two points. Potential difference is measured in volts (V). A volt is defined as the difference of potential that carries one ampere of current against a particular quantity of resistance.

Every battery has its potential difference printed on it. For example, a dry cell carries 1.5V.

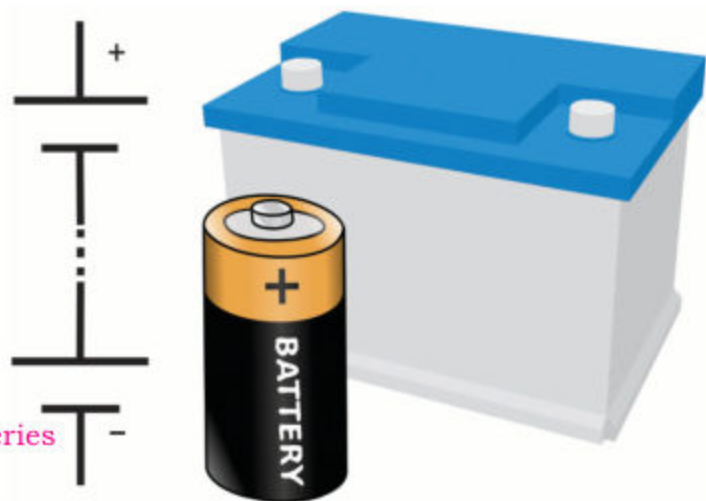


Figure 11.10 Electrical Batteries

RESISTANCE AS AN OPPOSITION TO THE FLOW OF CURRENT

- ✓ Explain the resistance as an opposition to the flow of current.

Electric current flows better through some materials than others. Resistance is the hindrance to the flow of current. The charges collide within the conductor (wire) numerous times, during their journey through the electric circuit. Consequently, there is the hindrance (resistance) to the flow of the current.

The resistance of a wire depends on length of the wire and thickness of the wire. Longer wires have more resistance than short wires and thin wires have more resistance than thick wires. The unit of resistance is ohm (Ω).

RELATIONSHIP BETWEEN VOLTAGE AND RESISTANCE

- ✓ Describe the relationship between voltage and resistance.

In 1827, a German scientist George Simon Ohm discovered relationship between the voltage and current in an electric circuit. It is called Ohm's Law.

The relationship between voltage and resistance is described by mathematical equation.

$$\text{Resistance} = \frac{\text{Voltage}}{\text{Current}} \quad R = \frac{V}{I} \quad \text{or} \quad V = IR$$

“Current through a conductor is directly proportional to the potential difference or voltage applied across the conductor and inversely proportional to the resistance”.

Resistance is a ratio of voltage to current. Above equation shows that resistance is equal to the voltage divided by the current.

MEASURING CURRENT, VOLTAGE AND RESISTANCE

- ✓ Measure current by using different devices.

Following devices are used to measure current, voltage and resistance of an electric circuit:

- Current Ammeter
- Voltage Voltmeter
- Resistance Ohm meter



Figure 11.11 An ammeter, voltmeter and ohm meter to measure the electric current

An ammeter is the device to measure the amount of current in an electric circuit (figure.11.11). It is connected to the circuit in series so that full current passes through it. An ammeter does not change the amount of the current in a circuit because it has very low resistance.

HAZARDS OF ELECTRICITY

- ✓ List electrical hazards and precautionary measures to ensure the safe use of electricity at home.
- ✓ Describe why electricity is dangerous to human.

1. Electric shocks and electrocution:

Mostly unawareness is the main cause of electrical hazards. Following hazards cause injury or death.

- When a person touches the open plug or live wires, current flows through his body causing an electric shock to the person.
- The person is injured or may be killed.
- It can be prevented by using good quality electrical wires, that avoids its hazards.

2. Short Circuit:

- The large current flowing through the wires may cause the wire to overheat and a fire may start. Sometimes electrical sparks from a short circuit may also ignite flammable materials nearby and start a fire.
- To avoid this hazard do not connect too many electrical appliances to the same output socket.
- Do not forget to switch off an appliance as may also cause overheating of wires, which may cause a fire.

3. Safety devices:

We may safely use electricity by using following devices.

- Fuses.
- Miniature circuit breakers (MCBs)
- Earth wires.
- Three pin plug.
- Earth leakage circuit breaker (ELCB).

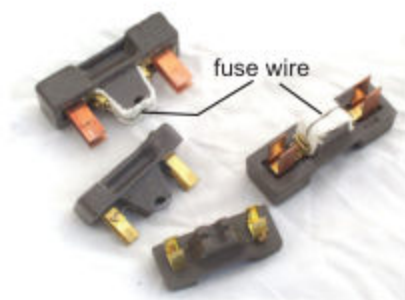


Figure 11.12 Fuses



Figure 11.13 Miniature circuit breakers (MCBs)



Figure 11.14 A Three pin plug



Figure 11.15 An earth leakage circuit breaker (ELCB)

DO YOU KNOW?

Heating effect of electricity can be dangerous. Many house and office fires have been caused by overheated faulty electrical wires or appliances.

Why is Electricity Dangerous?

Electric shock is highly dangerous and painful.

What can happen as a result?

- Muscles tighten up, making it almost impossible to pull away from the circuit.
- Lungs contract, making it hard to breathe.
- Heartbeat is interrupted and blood vessels tighten.
- Burns occur where the electricity enters and leaves the body.

This happens because the human body is a good conductor of electricity. Electricity flows easily through our bodies, because human body comprises 70 percent of water.

Secondly, that electricity always tries to find the easiest path to the ground.

If the electricity is strong enough, it can cause the victim's muscles to tighten up so much he or she can't let go.

Safety measures:

However, if you follow the following safety rules, you can use electricity without getting hurt.

1. Do not insert any metal object into electric sockets.
2. Do not touch a fallen electric wire, especially from power current lines.
3. Never touch electrical appliances with wet hands.
4. Make sure to not overload power sockets.
5. Never grab on to anyone who is shocked. Use a non-metallic object such as rubber or dry wood to shift the victim away from the electric wire.

Summary

- The movement of free electrons is called electric current.
- Electric current is measured in ampere (A).
- There are two types of circuit; series and parallel.
- The circuits in most businesses and homes are connected in parallel.
- Energy is the ability to do work and electricity is a kind of energy.
- Different household appliances are used for the conversion of electrical energy into other forms of energy.
- Potential difference between two points in a circuit is called voltage.
- Resistance is the hindrance to the flow of current.
- Safety devices can be used to prevent electricity hazards.
- Human body is good conductor of electricity, because it contains large amount of water and other liquids.

Review Exercises

1. Complete each of the following sentences by writing the correct term:

- i) The circuit that provides only one path for the current is called _____.
- ii) One of the unit on our electricity meter is known as _____.
- iii) The unit of potential difference is _____.
- iv) The unit of electric current is _____.
- v) The unit of resistance is _____.

2. Give short answers:

- i) What is an electric current?
- ii) What causes the current to flow in an electrical circuit?
- iii) What is the main difference between a series circuit and a parallel circuit?
- iv) Is the lighting system in your classroom a parallel or series circuit? Give reason of your answer.

3. Explain hazards of electricity.

4. Suggest some ways to safe use of electricity.

PROJECT

An electric current from lemon/potato

What I need:

- A lemon
- About 5 cm long piece of copper wire
- 1 meter long insulated wire with bare ends
- A galvanized nail
- A compass

What to do:

- Press the lemon hard on a table to make it juicy inside.
- Wind the plastic coated wire round the compass several times.
- Twist one of the bare ends of wire around the copper wire
- Push the copper wire into the lemon.
- Wind the other bare end of the wire around the nail. Push the nail into the lemon, about 3 cm from the copper wire.
- Now look at the compass needle and record your observations.



Circuit with Lemon

What I observed:

Activity Questions:

- Did you observe any change? yes/ no. Give reason of your observation?
- What is the role of copper wire; the nail and the lemon juice in this activity?
- Investigate if you use potato instead of a lemon, does this make electricity too?

What I conclude: